An Investigation of PDE and Network Models for Vegetation Patterns

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Abstract

We attempt to synthesise different approaches to modelling vegetation patterns, namely classical models using advection-diffusion continuous PDE systems and cellular discrete time models. We first introduce the classical Klausmeier model and analyse its bifurcation structure, as well as extensions of the model and the effects of these on the steady states and pattern forming. We then shift our attention to the more sophisticated Stewart et al. model, where we analyse the model's behaviours through numerical simulations in MATLAB using the provided code. We investigate the effects of varying levels of constant rainfall on the models, with Klausmeier exhibiting multi stability and pattern forming instabilities under spatial perturbations, whereas the Stewart model exhibits strong path-dependence and long-term oscillatory behaviour in biomass time series data. This comparison allows us to assess the extent to which the more mathematically tractable models, like the Klausmeier model, can reproduce the effects and behaviours exhibited in the more detailed Stewart model.

1 Introduction

Semi-arid conditions are prone to many undesirable ecological phenomena. Such regions are exposed to minimal yet incredibly variable rainfall patterns which can lead to desertification, amongst other phenomena. This means it is important for us to be able to predict what factors can cause desertification to occur, so that we can understand what measures are needed to prevent or control such events. Modelling these ecosystems allows us to simulate these conditions over time and see what measures are appropriate to improve and manage ecosystem resilience, before said ecosystems are eventually led to collapse. The two emerging methods for modelling these ecosystems are continuum methods, such as classical advection-diffusion PDE models, and discrete network models.

PDE models are generally simplified to ensure that they are mathematically tractable and pleasant to analyse. The Klausmeier model for vegetation patterns [1] uses a PDE system, which is purposefully kept simple so that we can apply well-known tools, such as linear stability analyses, as well as analysing their behaviours with computational tools and simulations without the need for HPCs. Instead we can discretise these systems and put them into ODE solvers in software like Python/MATLAB etc. PDE models for vegetation patterning tend to capture core motifs of the system, and build the model around them, whilst simplifying other factors.

Network models for vegetation tend to be mathematically and computationally complex and can sometimes require extensive computation, however are much more sophisticated and often mathematically intractable due to the increased complexity of the model. They aim to be as realistic as possible. Models like the Stewart et al. [2] require heavy amounts of simulation to analyse and identify key effects and motifs in the system, and as such are much more modern.

In this project, I attempt to explore the different approaches of modelling vegetation patterns with these two approaches by exploring their dynamics and how their different underlying assumptions can affect core tendencies in the models.

2 The Klausmeier Model

The Klausmeier Model uses a system of PDEs to model vegetation biomass and water in semi-arid conditions.

$$\frac{\partial w}{\partial t} = A - w - wb^2 + \nu \frac{\partial w}{\partial x} + D_w \nabla^2 w,$$
$$\frac{\partial b}{\partial t} = wb^2 - mb + D_b \nabla^2 b. \tag{1}$$

For the basic model, w represents surface water and b is our biomass. The remaining parameters are described below.

Parameter	Interpretation
A	Constant rainfall input into the domain.
ν	Strength of downslope water transport.
D_w	Water diffusion coefficient.
D_b	Biomass diffusion coefficient.
m	Vegetation mortality rate.

Table 1: Parameters in the non-dimensional Klausmeier model.

Water moves down the slope, and vegetation depletes local water availability through non linear uptake. This results in the bands migrating uphill, albeit incredibly slowly. Much of the literature surrounding Klausmeier and other models considers this positive feedback loop and local water infiltration, sufficient mechanisms for pattern formation, with other factors (grazing, slope, herbivory, etc.) not being considered essential. However, these extra factors can determine the parameter ranges for which pattern formation can occur [3]
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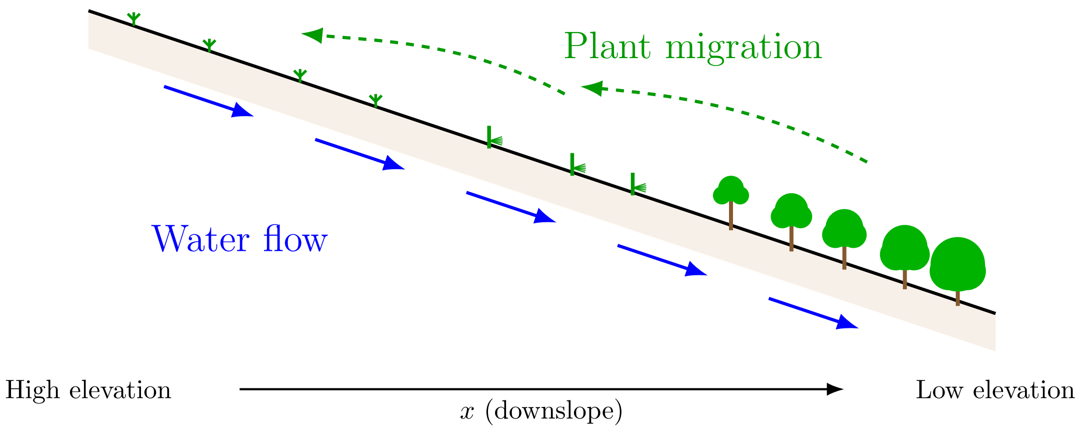


Figure 1: Illustration of Band Migration and Water Flow Positive Feedback Loop

Klausmeier is a purposefully simplified model, and omits some key ecological factors, including grazers, hydrodynamics beneath the surface, variable rainfall amongst others.

2.1 Linear Stability Analysis of Klausmeier

2.1.1 Homogeneous Equilibria and Stability

We find homogeneous equilibria by setting all partial derivatives to zero, giving

$$A - w - wb^2 = 0,$$
$$wb^2 - mb = 0.$$

From there we can solve these equations, first giving us our bare-soil equilibrium $(w_0, b_0) = (A, 0)$, which is permissible for $A > 0$.

We can then consider $b \neq 0$ with $w_* = m/b_*$, giving us our vegetated equilibria from the quadratic $mb^2 - Ab + m = 0$, which we can then solve to give

$$w_\mp = \frac{A \mp \sqrt{A^2 - 4m^2}}{2} \qquad b_\pm = \frac{A \pm \sqrt{A^2 - 4m^2}}{2m}.$$

These exist when the argument in the square root is positive, thus $A > 2m$ with $m > 0$. If $A > 2m$ then we have 3 equilibria, if $A = 2m$ then we have 2 equilibria and if $A < 2m$ then we have only the bare-soil equilibrium.

Next we can consider the stability of our equilibria and the conditions of stability. We start by applying the Trace-Determinant criterion on the homogeneous system's Jacobian.

Equilibrium (w, b)	**Interpretation**	**Stability and conditions**
$(A, 0)$	Bare soil Equilibrium	Stable Everywhere
(w_-, b_+)	High Biomass Vegetated Equilibrium	Stable if $b_+^2 > m - 1$ and $A > 2m$
(w_+, b_-)	Low Biomass Vegetated Equilibrium	Unstable everywhere if $A > 2m$
$(w_\mp, b_\pm) = (1, 1)$ if $A = 2m$	Uniform Vegetated Equilibria	Saddle Node, so not asymptotically stable

Table 2: Stability Criterion of Classical Klausmeier

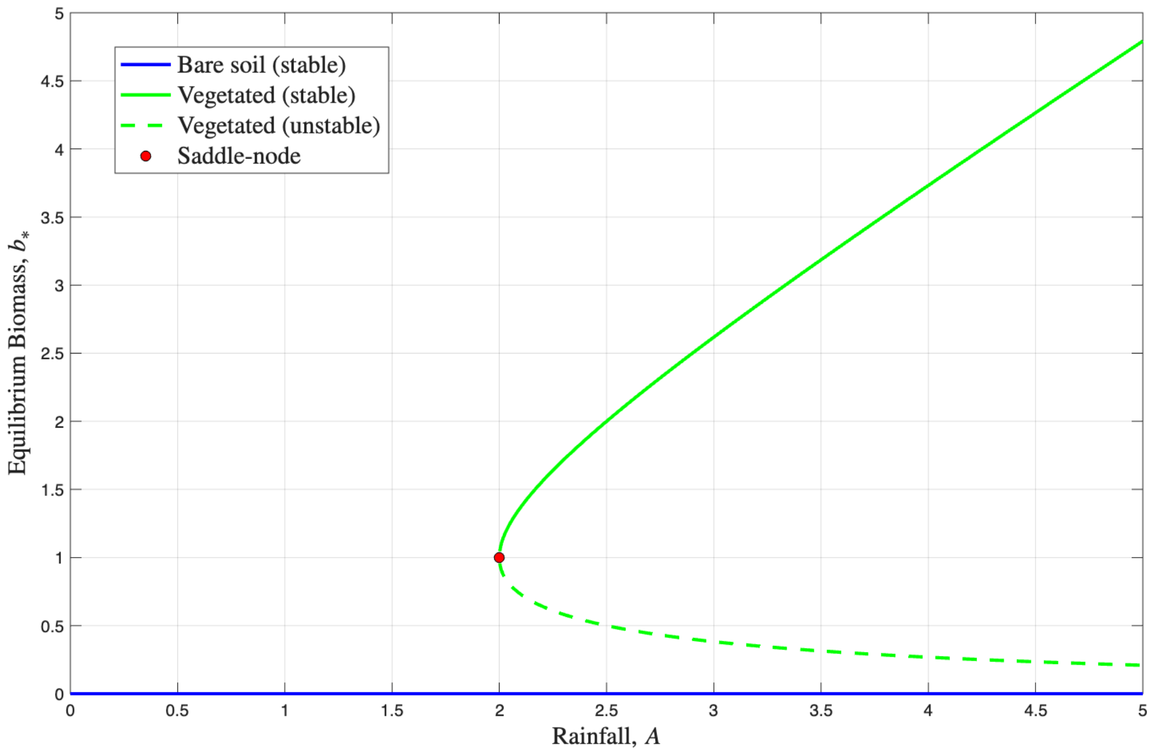


Figure 2: Bifurcation Diagram of Spatially-Homogeneous Klausmeier Model, with $m = 1$. Represents the equilibria discussed above over different values of A, with the stability at those values.

2.1.2 Stability under Spatial Perturbations

We begin by taking the homogeneous equilibria and perturbing them, i.e.

$$w = w_* + \epsilon W(x,t),$$
$$b = b_* + \epsilon B(x,t).$$

for some $\epsilon \ll 1$. We then substitute these into the full PDE system, and neglect terms with $\mathcal{O}(\epsilon^2)$ to get the linearised equations

$$\begin{aligned}
\frac{\partial W}{\partial t} &= -W - 2w_* b_* B - b_*^2 W + \nu \frac{\partial W}{\partial x} + D_w \nabla^2 W, \\
\frac{\partial B}{\partial t} &= 2w_* b_* B + b_*^2 W - mB + D_b \nabla^2 B.
\end{aligned}$$

We can then write our matrix equation as

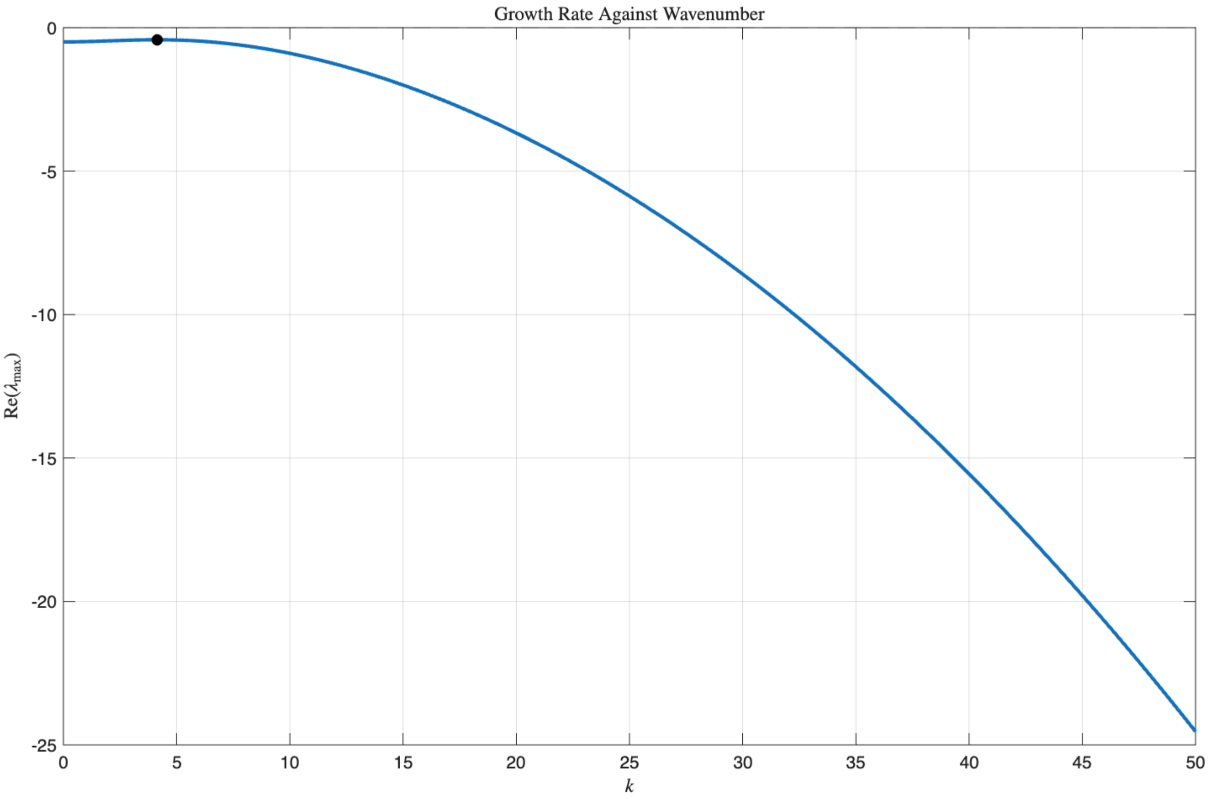
$$\frac{\partial}{\partial t}\begin{pmatrix} W \\ B \end{pmatrix} = \begin{pmatrix} -(1+b_*^2) & -2w_* b_* \\ b_*^2 & 2w_* b_* - m \end{pmatrix}\begin{pmatrix} W \\ B \end{pmatrix} + \begin{pmatrix} \nu W_x + D_w \nabla^2 W \\ D_b \nabla^2 B \end{pmatrix}$$

and introduce an ansatz of $(W, B) \propto e^{\lambda t + ikx}$.

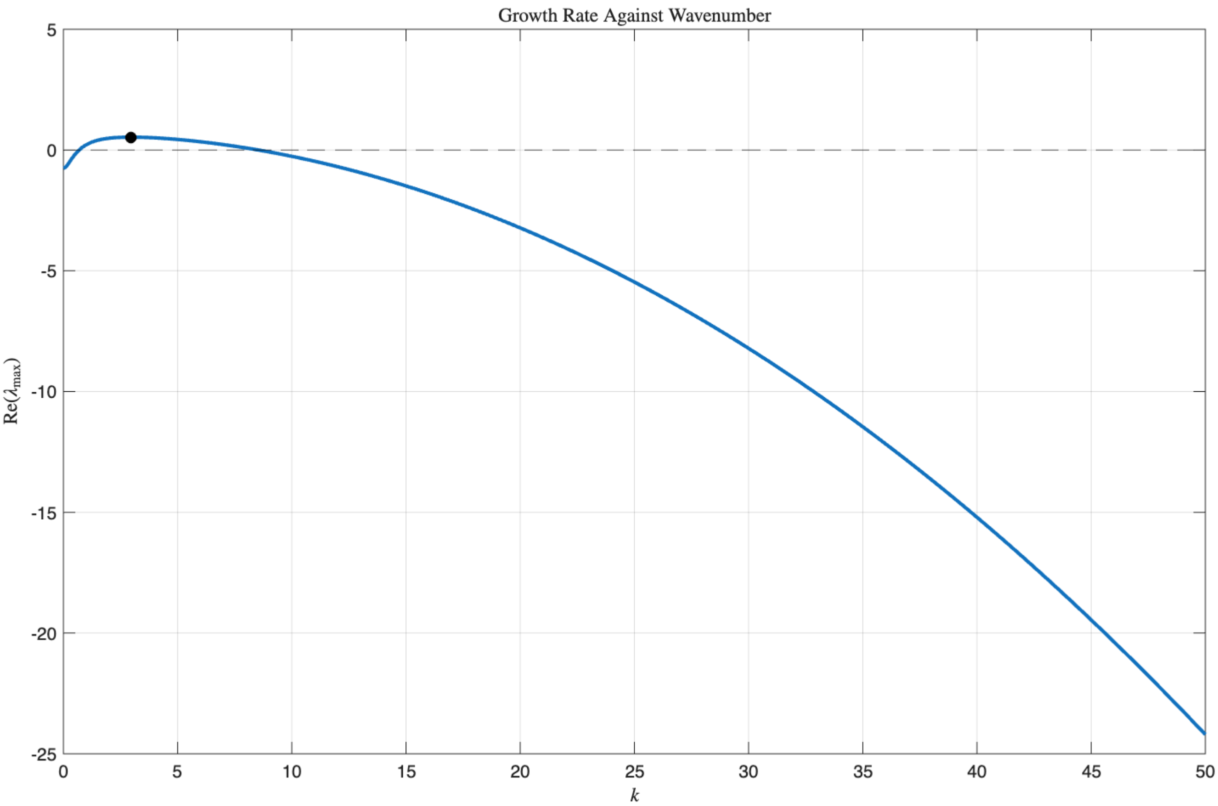
Once we substitute the ansatz into our linearised equation we can derive the eigen-equation as

$$\lambda \begin{pmatrix} W \\ B \end{pmatrix} = \underbrace{\begin{pmatrix} -(1+b_*^2) - D_w k^2 + i\nu k & -2w_* b_* \\ b_*^2 & 2w_* b_* - m - D_b k^2 \end{pmatrix}}_{M} \begin{pmatrix} W \\ B \end{pmatrix}.$$

We use $\lambda(k) = \frac{\mathrm{Tr}(M) \pm \sqrt{\mathrm{Tr}(M)^2 - 4\det(M)}}{2}$ to find the growth rates, as well as the wave numbers that maximise the growth rate for each parameter set. We can then find the most unstable wave number to determine spatial stability and whether or not pattern formation is likely to occur for different parameter sets. This allows us to compare theoretical wave speeds with the wave speeds in the simulations.



(a) $A = 5, m = 0.5, D_b = 0.01, D_w = 1, \nu = 10$. The perturbations do not result in pattern formation as the real part of the growth rate remains negative for all k, thus homogeneous equilibrium is linearly stable to the perturbations.



(b) $A = 2.5, m = 0.8, D_b = 0.01, D_w = 1, \nu = 10$. The system starts in a stable equilibrium, and spatial perturbations push the system to be unstable, with a most unstable wave number around 3.

Figure 3: Dispersion Curves for different parameter sets of the Klausmeier model.

3 Extensions of the Klausmeier Model

The Klausmeier model is comprised of advection, diffusion, rainfall, and simple plant mortality and uptake terms. We can introduce other terms, such as wind, grazing and subsurface water dynamics amongst others, which can change the parameter sets required for vegetation bands.

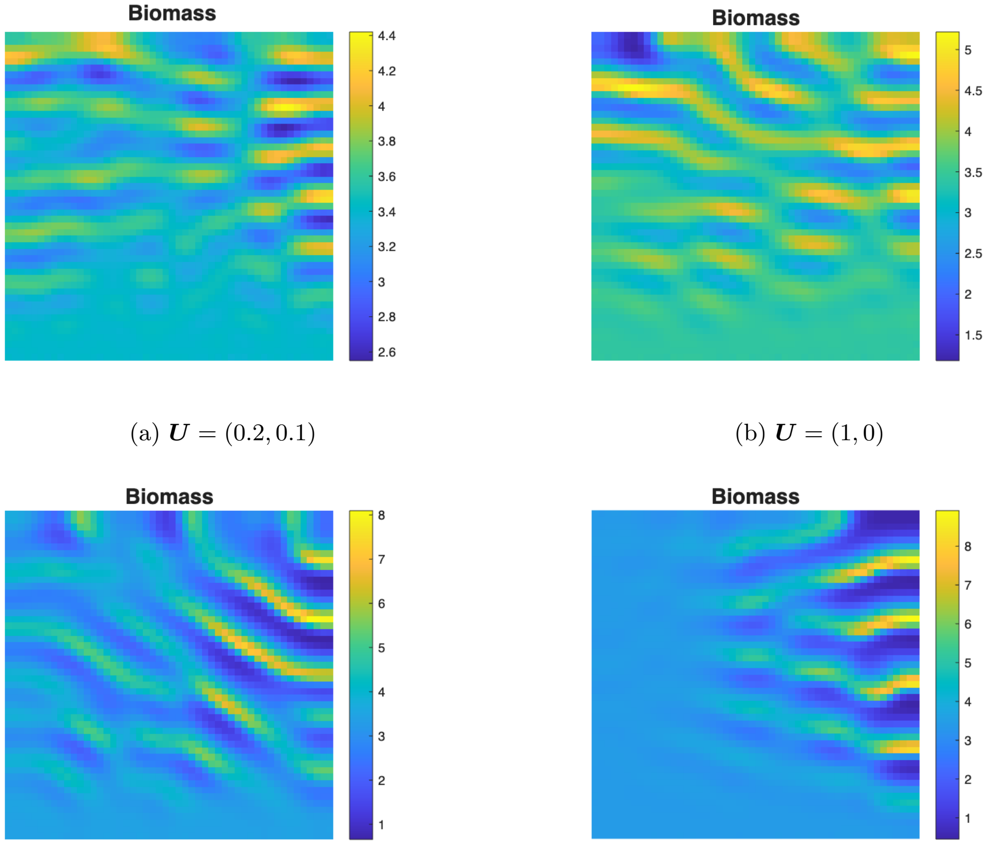
3.1 Wind

Wind affects plant biomass dynamics via seed dispersal as well as potential evaporation of water (however this is small enough of an effect to be considered insignificant). Wind, when strong enough, can also affect plant mortality. The simplest implementation of this would be an extra linear mortality term; however, it is easier to combine that with m in our current version.

Other ways of introducing wind are through advective terms for water, and a simple death term for biomass (which can be factored into m). This also means that greater control over the orientation of the patterns is allowed. We can do this by introducing a vector field $\boldsymbol{U}$ to the system, so the water equation becomes

$$\frac{\partial w}{\partial t} = A - w - wb^2 + \nu \frac{\partial w}{\partial x} - \boldsymbol{U} \cdot \nabla w + D_w \nabla^2 w. \tag{2}$$

This also allows the direction of water transport to be controlled through another parameter. Some example simulations, alongside the vector field used are given below.



(a) $\boldsymbol{U} = (0.2, 0.1)$

(b) $\boldsymbol{U} = (1, 0)$

(c) $\boldsymbol{U} = (2.5, 0)$

(d) $\boldsymbol{U} = (-2(y - \frac{L}{2}), 2(x - \frac{L}{2}))$

Figure 4: Simulations of the wind-extended Klausmeier model with different vector field wind terms. All simulations were run with $A = 2, m = 0.54, \nu = 5, D_w = 1, D_b = 0.01$.

The limitation of this extension is that it is simply adding an extra advective term which does not directly create or destroy water, which in certain windier environments can be useful, but does not act as an essential term for explaining the specific patterns that arise. However, when considering the danger of desertification in these ecosystems, considering plant mortality is a potentially more useful extension than the extra advective control of water.

3.2 Grazing

Many ways of introducing grazers have been documented. The simplest way is by introducing an extra linear mortality term that we can use to control mortality separately. We can also introduce a Holling-type functional response term in the biomass equation, $\frac{gb}{k+b}$, as an extra mortality term.

Another thing we can do is introduce a third variable, say $g(x,t)$, and use a third advection diffusion equation to model the population dynamics of the grazers, before incorporating the variable into the equations for water and biomass. Introducing the Holling-type term into the third equation then gives

$$\begin{aligned}\frac{\partial w}{\partial t} &= A - w - wb^2 + \nu\frac{\partial w}{\partial x} + D_w\nabla^2 w,\\ \frac{\partial b}{\partial t} &= wb^2 - mb - \frac{\alpha g b}{k+b} + D_b\nabla^2 b,\\ \frac{\partial g}{\partial t} &= \frac{\alpha\eta g b}{k+b} - \mu g + D_g\nabla^2 g,\end{aligned} \tag{3}$$

where α is the maximum grazing rate, k is a saturation constant, η is a term representing how efficiently grazing produces new grazers (birth rate), μ is a simple grazer mortality term and D_g is a diffusion coefficient.

We can also add a taxis term $-\chi\nabla\cdot(g\nabla b)$ into the grazer equation allowing the grazers to move towards high plant biomass regions. This adds more realism and less randomness to the grazer dynamics, but makes the model slightly slower to run due to the extra derivatives, on top of the 5 extra parameters we've added with the extra PDE.

3.3 Issues with extending the Klausmeier Model

Whilst extending this relatively simple model seems appealing, as we can keep the classical advection-diffusion PDE structure whilst introducing factors that were initially omitted by Klausmeier, it often results in the mathematical tractability being compromised.

Other models like the Rietkerk model [3] run into a similar issue of making the model much more computationally expensive, with parameter identifiability becoming an issue due to the many extra parameters used in the extra PDE.

These additions do allow us to see how pattern formation changes with their inclusion, which can be more representative of certain vegetation patterns in different environments. Moreover, models like Rietkerk act almost as a bridge between much more simplified models like Klausmeier, and more sophisticated models like Stewart, and allows us to choose whether a trade off of simplicity versus realism is worth it when analysing pattern formation.

4 The Stewart et al. Model

4.1 Introduction

The Stewart Model is a network-based model for dryland ecosystems proposed in 2014. It works by describing the landscape as a lattice of size $N \times N$ and using a set of discrete time step difference equations to describe a network. Resources and propagules are then distributed through local ecological processes via vectors. The conceptual outline of the system is given below.

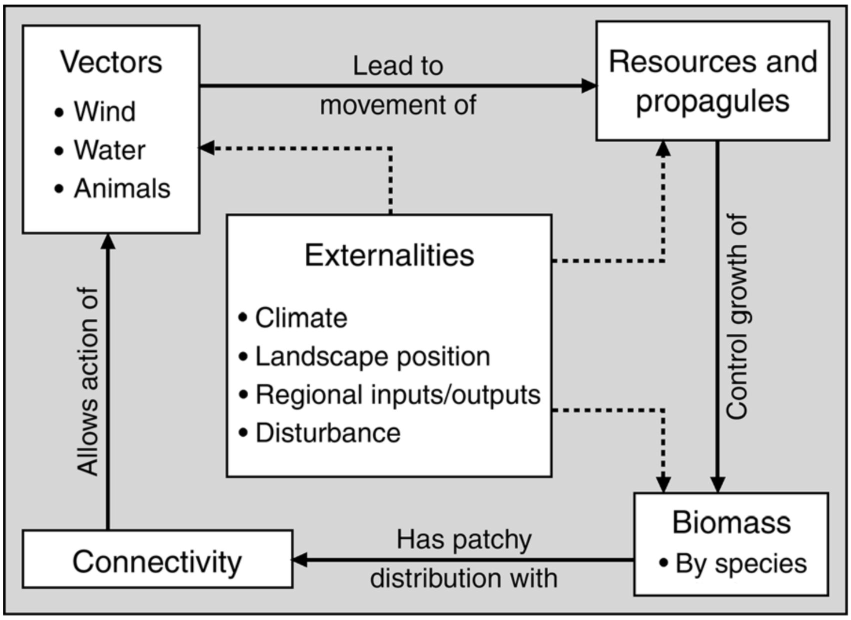


Figure 5: Flow chart describing the dynamic relationship between different elements of the model. Solid lines represent the modelled processed, and dashed lines represent the externalities which affect those processes.

A fundamental contrast between the two models is that in the Stewart model, the state variables are updated locally with discrete time steps, however the Klausmeier model uses continuous fields governed by PDEs.

The model stores vegetation biomass and, with each time step, determines the required resources needed for plant growth through a maintenance term M. Through subroutines, the model first redistributes spatially using connectivity matrices, before using resource availability terms to update local vegetation dynamics, such as growth and mortality. There are many other key differences in the underlying assumptions used in both Klausmeier and Stewart, which are further outlined in the original paper [2] .

Another addition of the Stewart model is the addition of nitrogen movement. This adds another potentially limiting factor of biomass growth which can result in some interesting effects. In simulations I found that increasing constant rainfall in the model does not always result in unlimited growth in biomass, and as water grows abundant, nitrogen becomes the limiting resource.

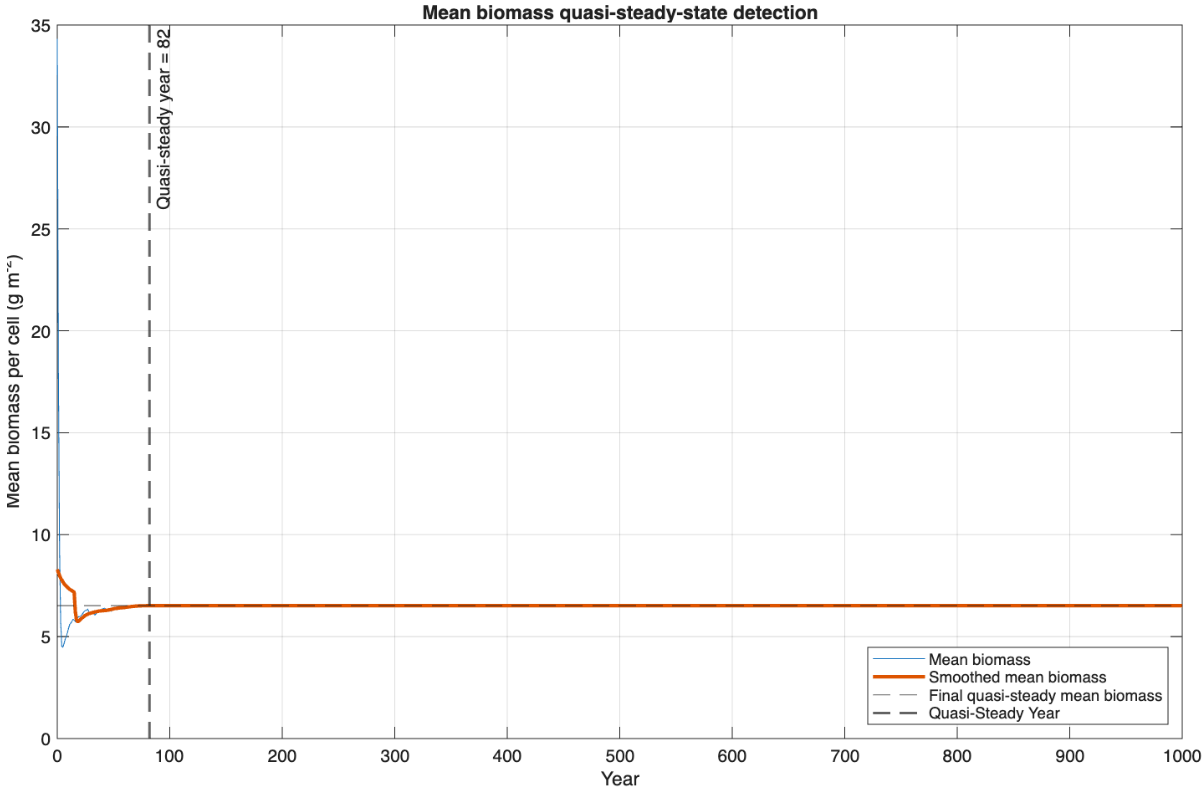
Moreover, the model tracks the movement of two species, rather than just a total biomass like in Klausmeier. We instead have black grama and creosotebush, This allows us to additionally track community composition of biomass, which can result in some interesting dynamics.

The model also allows rainfall data to be variable rather than constant. As we are analysing the model's signatures in comparison to Klausmeier however, it is more useful for us to use constant rainfall for now.

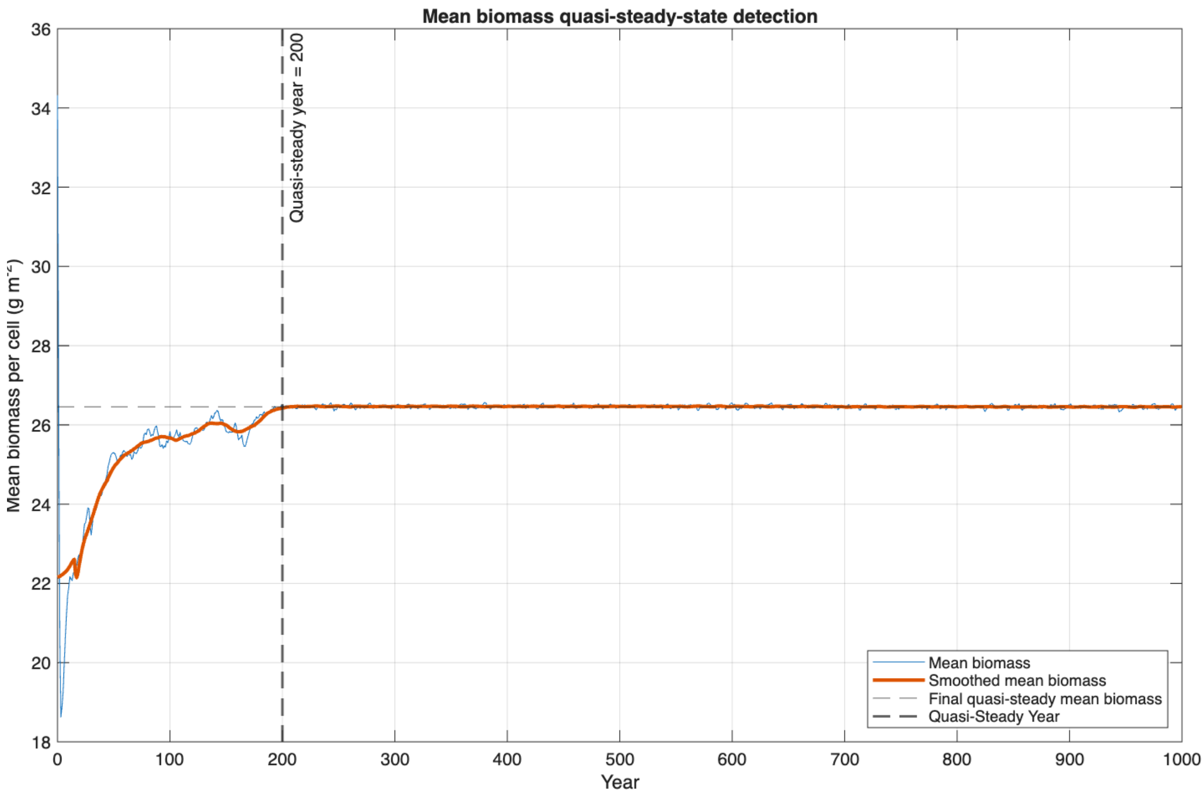
4.2 Quasi-Steady States for Biomass

Steady states in a model like this are often analytically intractable due to the intricacy of the model. This means that studying asymptotic behaviour should be done via simulation of the model over lots of time steps. Unlike the Klausmeier model, steady states are not always to be expected, and the model often reaches oscillatory regimes. Using the Stewart code and MATLAB, I created a programme which allows us to see if and when the biomass first reaches a quasi-steady state.

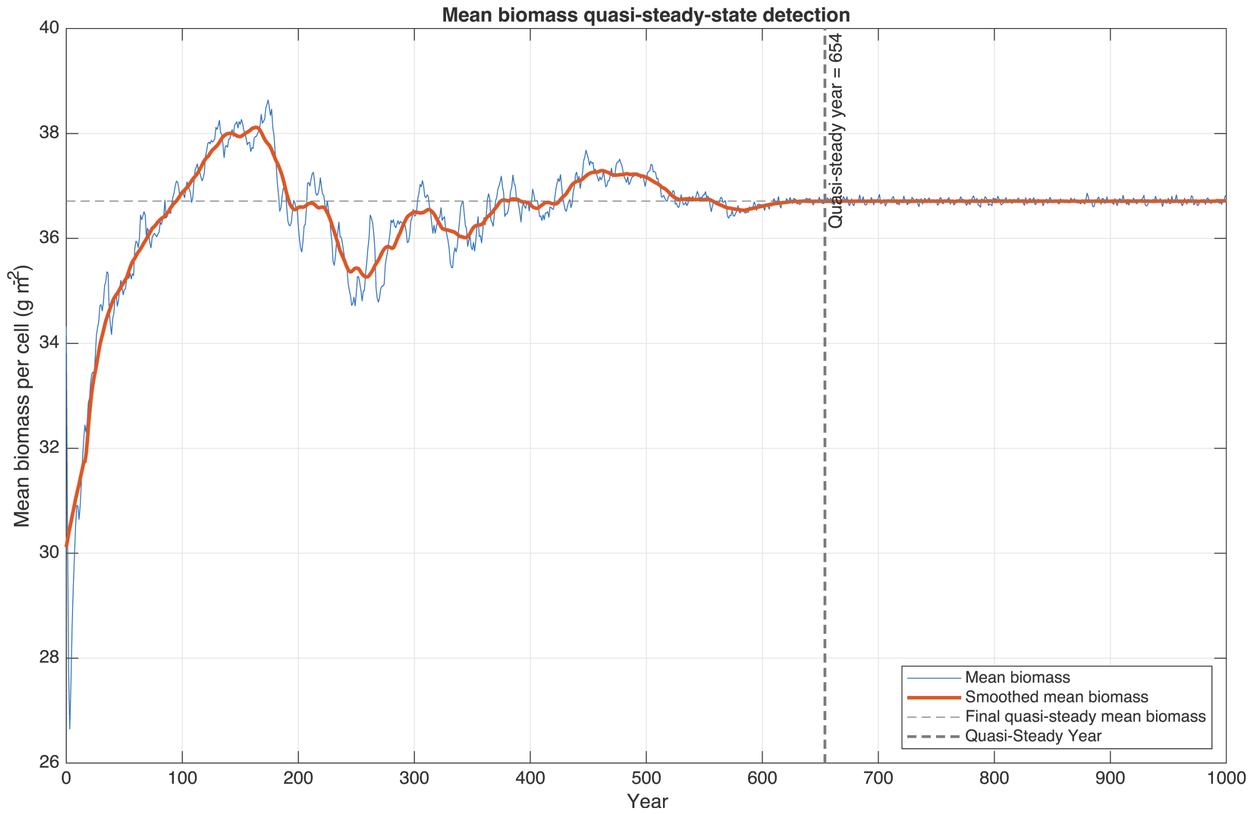
This was achieved by running the original code with constant rainfall, and storing the biomass entry at each time-step, and testing if the biomass stays in a suitable range of tolerance, i.e checking that $|B_{t+1} - B_t| < \epsilon$ over a set number of years.



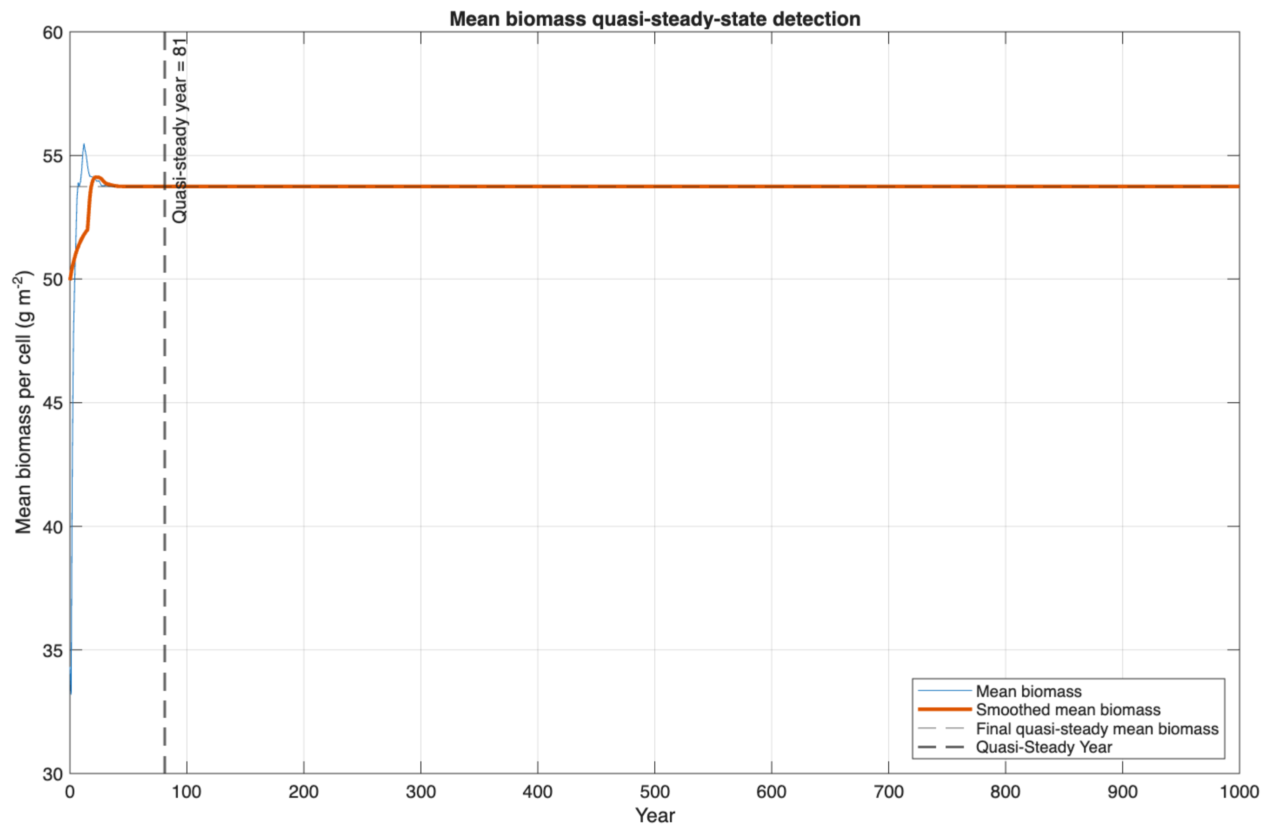
(a) Constant rainfall of 25 mm.



(b) Constant rainfall of 100 mm.



(c) Constant rainfall of 150 mm.



(d) Constant rainfall of 300 mm.

Figure 6: Mean biomass dynamics and detection of the quasi-steady regime under different levels of constant rainfall.

Our simulations seem to show that the larger the constant rainfall, the longer the system seems to take to reach a quasi-steady state regime, however, once constant rainfall gets too big, the system once again stabilises earlier.

One possible explanation for this is that the higher rainfall permits greater growth rates of biomass, thus the system requires more time to settle and reach some form of equilibrium, however once the rainfall is too high, the nitrogen levels seem to become limiting, so the system fails to sustain growth, so reaches it's steady state quicker.

4.3 Effects of Drought

We can run simulations where during a 400 year period, perhaps years 50-100 are run with rainfall of 25mm yr^{-1}, and the rest with normal rainfall. Whilst not realistic in reality, with droughts in most environments lasting around 3-6 years [4], running these simulations is important as it allows us to analyse the robustness of the model to low rainfall, and whether we are actually able to predict desertification through the model when there is a complete lack of resources to sustain the biomass.

We find that the model tends to be quite robust to long periods of low rainfall, and is seemingly able to recover really well every time, with biomass seemingly able to survive almost anything. We also find that often the biomass actually grows higher than the original amount. This should be considered a limitation of the Stewart model, as drought periods of this length would likely result in desertification. More precisely, the biomass seems to be able to persist at a low quasi steady state during drought periods, leaving enough biomass to allow for recovery when the drought ends.

A key reason for this level of robustness is the way in which plant mortality is calculated. As mentioned in section 4.1, plant mortality is calculated via a maintenance term for water and nitrogen. We calculate biomass at the next step in 3 different cases, one where neither resource is limiting, one where water is and one where nitrogen is. When water is limiting, we use

$$B_{t+1} = B_t \frac{d_i}{M_{i,w}}(1 - \mu_i), \tag{4}$$

and where nitrogen is limiting we use

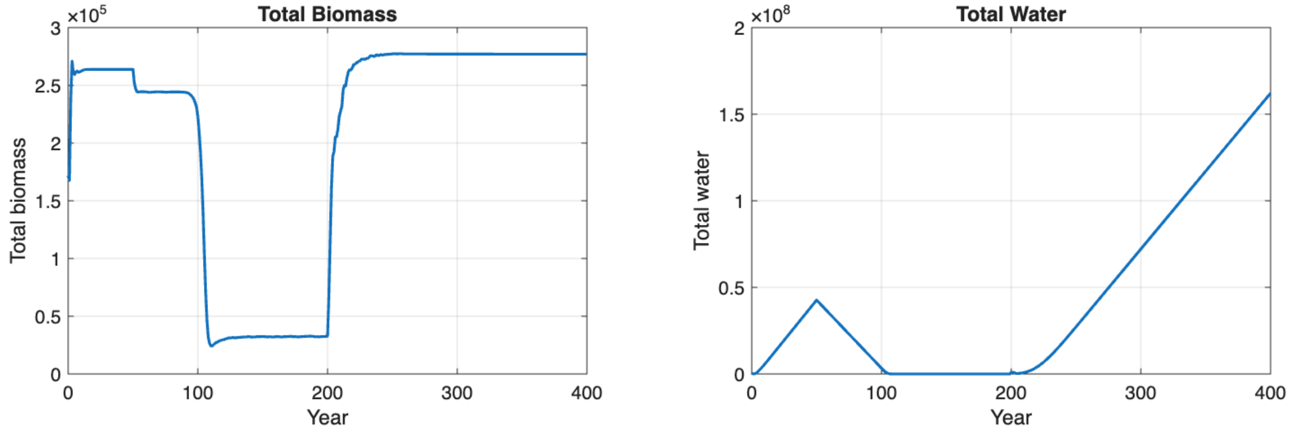
$$B_{t+1} = B_t \frac{N_i}{M_{i,N}}(1 - \mu_i). \tag{5}$$

Nitrogen-limited biomass is determined by how much is actually available, N_i, whereas the water limited biomass is determined by d_i, which is a drought term that acts as a survival multiplier for the shrub and grass biomass. The key here is that this is all multiplicative, so biomass struggles to die off, but the drought modifier also reduces the biomass that is lost. We also consider the maintenance for water and nitrogen M_i, which is higher when there's more biomass in each cell, and much smaller when there isn't, which can contribute to the model being too robust in these situations.

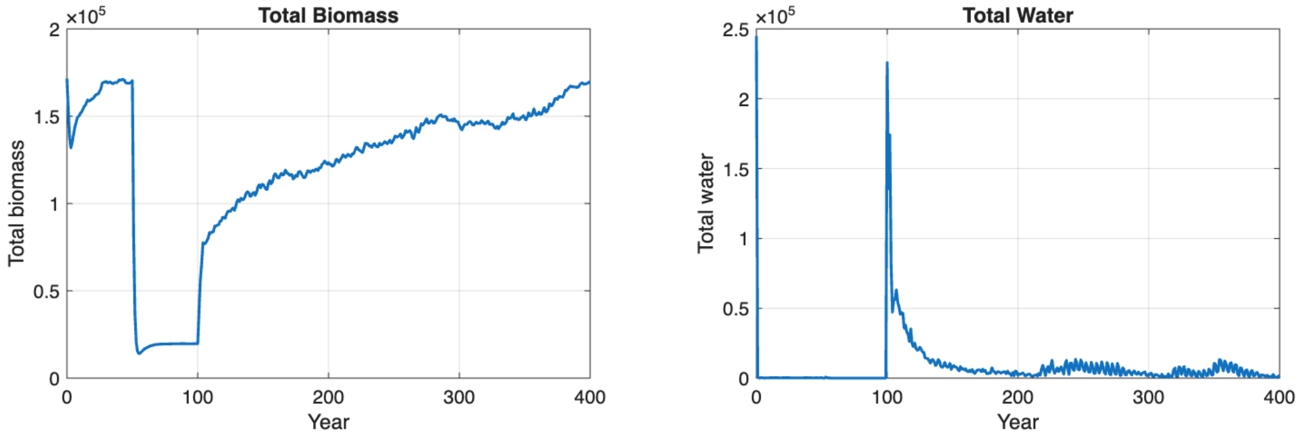
Overall, these drought simulations tell us that whilst the model can never fully reach 0 but can get very close due to the multiplicative nature of the update equations, the bigger issue stems from drought mortality rules of water-limited biomass, which allows biomass to survive even under lower water availability for extended time periods. Whilst recovery is normal in these kinds of environments, it's important that we are confident that the model is able to predict desertification efficiently if we are to figure out what factors cause such effects.

4.4 Continuation with Varying Rainfall and Path Dependence

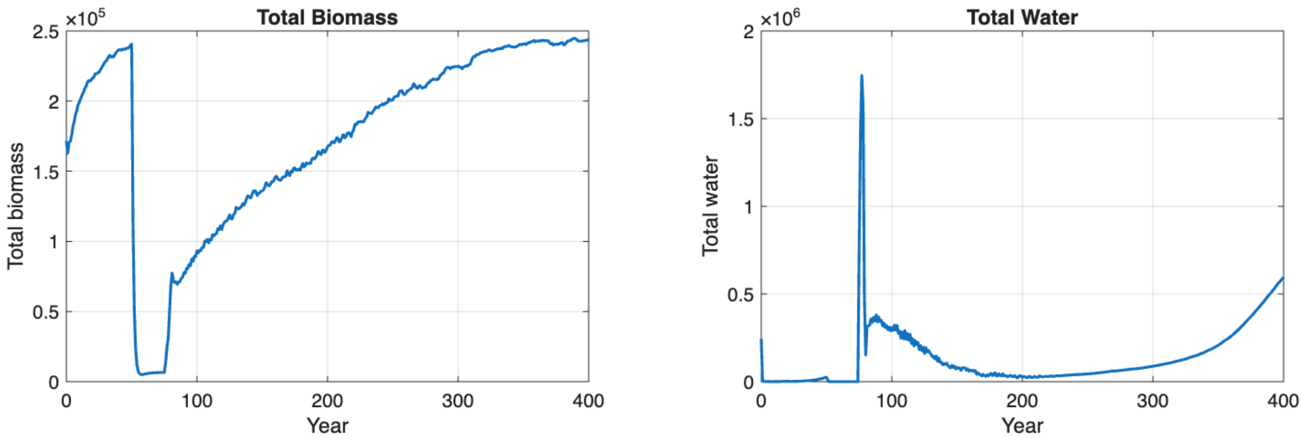
We can simulate the model over different levels of constant rainfall to see whether or not the model exhibits any interesting behaviours, similar to a bifurcation diagram for a continuous model. We do this via continuation, where we save the previous simulation's final state and initialise the next using said state. This allows us to consider what happens if we start from small rainfall versus what happens if we start from high rainfall.



(a) 400 mm/year rainfall with a 25mm/year drought between years 51 and 200.



(b) 150 mm/year rainfall with a 15mm/year drought between years 51 and 100.



(c) 200 mm/year rainfall with a 5mm/year drought between years 51 and 75.

Figure 7: Responses of the Stewart model to different drought scenarios.

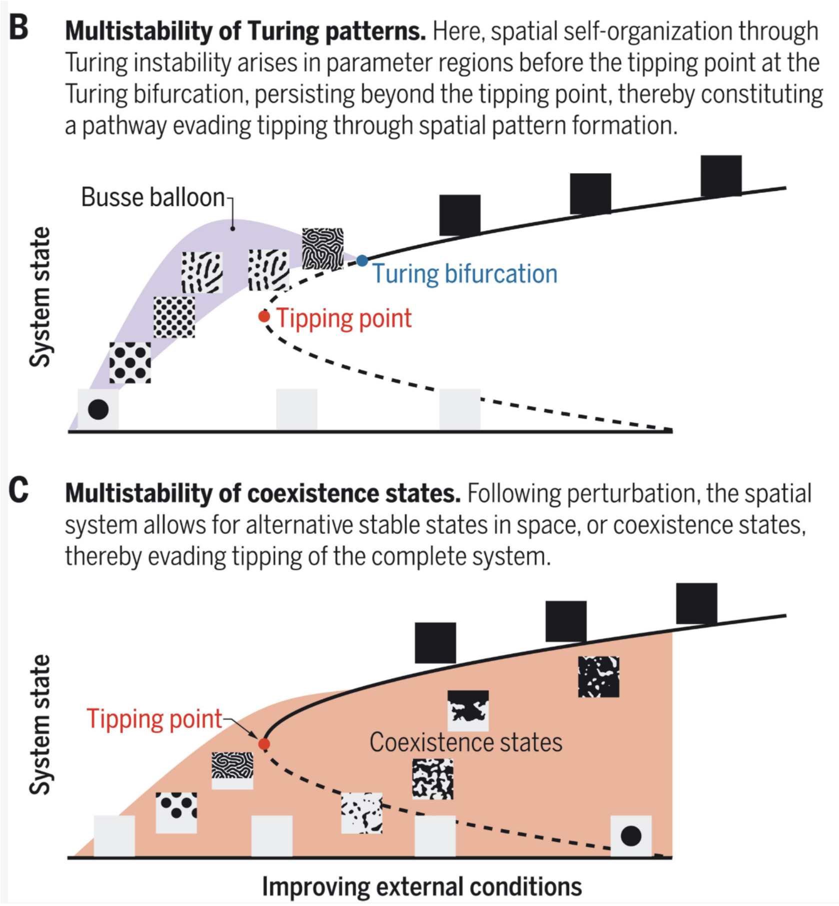
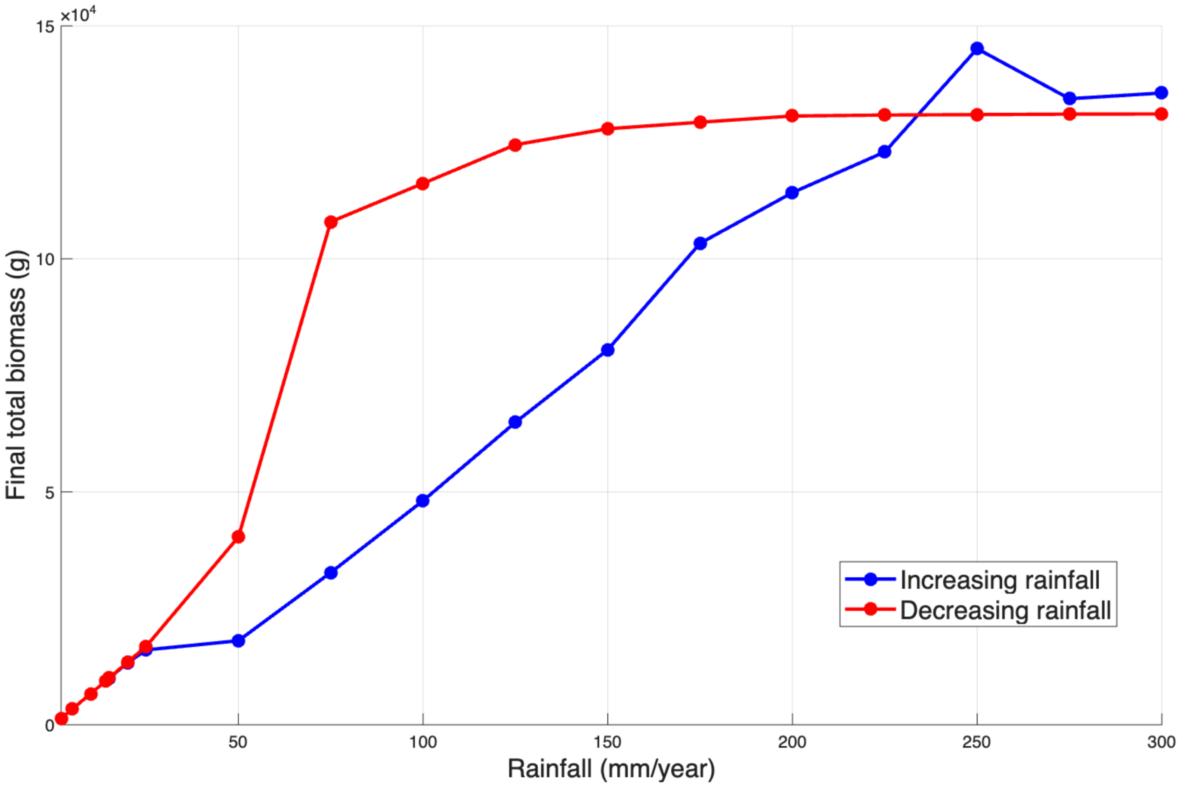
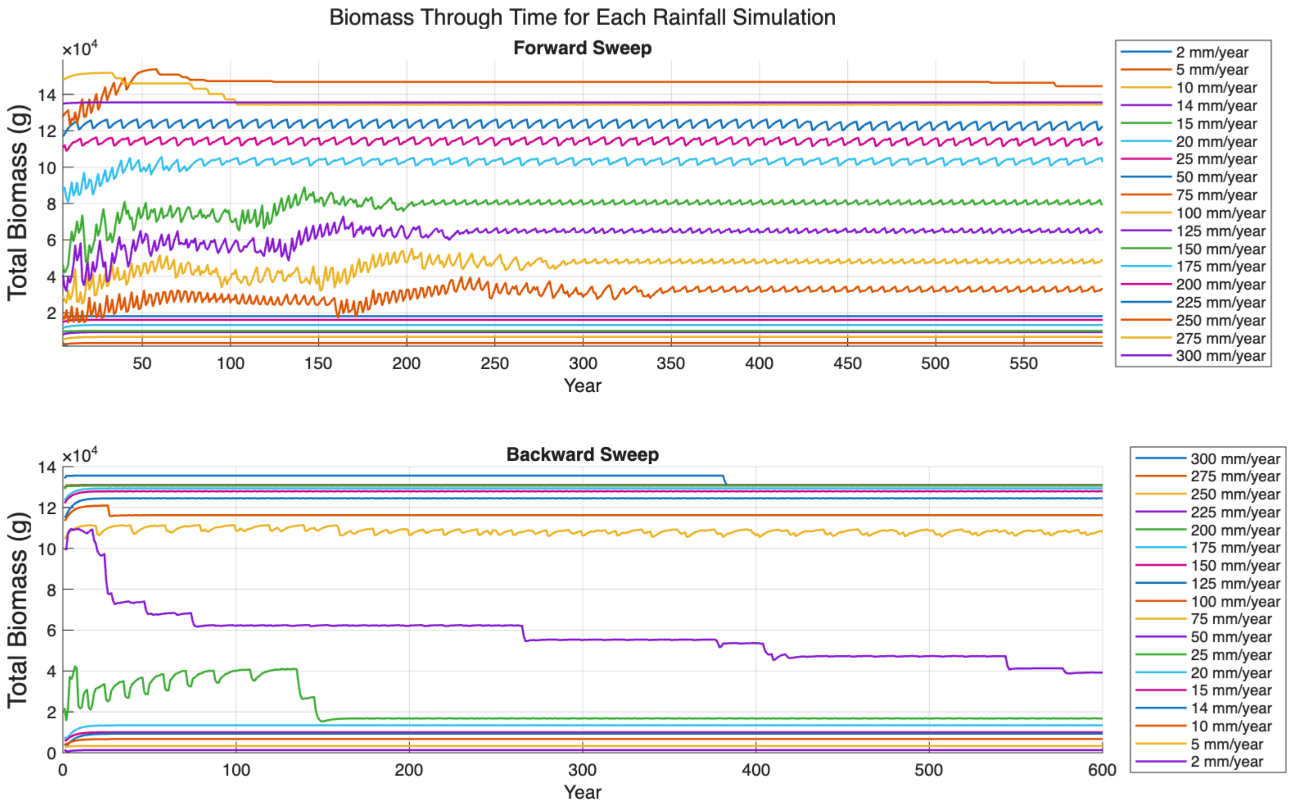


Figure 8: From Rietkerk, M., et al. "Evasion of tipping in complex systems through spatial pattern formation." Science 374.6564 (2021): eabj0359.

One change that is important for us to make in the implementation of the model is ensuring that the check for cells being un-vegetated is a threshold rather than a harsh zero condition. As discussed in 4.3, the biomass is extremely hard to kill off, so the biomass will rarely reach zero, however in simulations with extremely low rainfall, we end up with extremely small values ($\approx 10^{-300}$) which can result in NaN entries as we are dividing by something very close to 0, even though at that size, the cell should be considered un-vegetated anyway.



(a) Continuation diagram.



(b) Biomass time series.

Figure 9: Continuation signatures of the Stewart model. 600 year simulations at a range of constant rainfall values. Run with a field size of 50×50, upwards nitrogen flux of 1.5, and other input parameters found in the original paper [2].

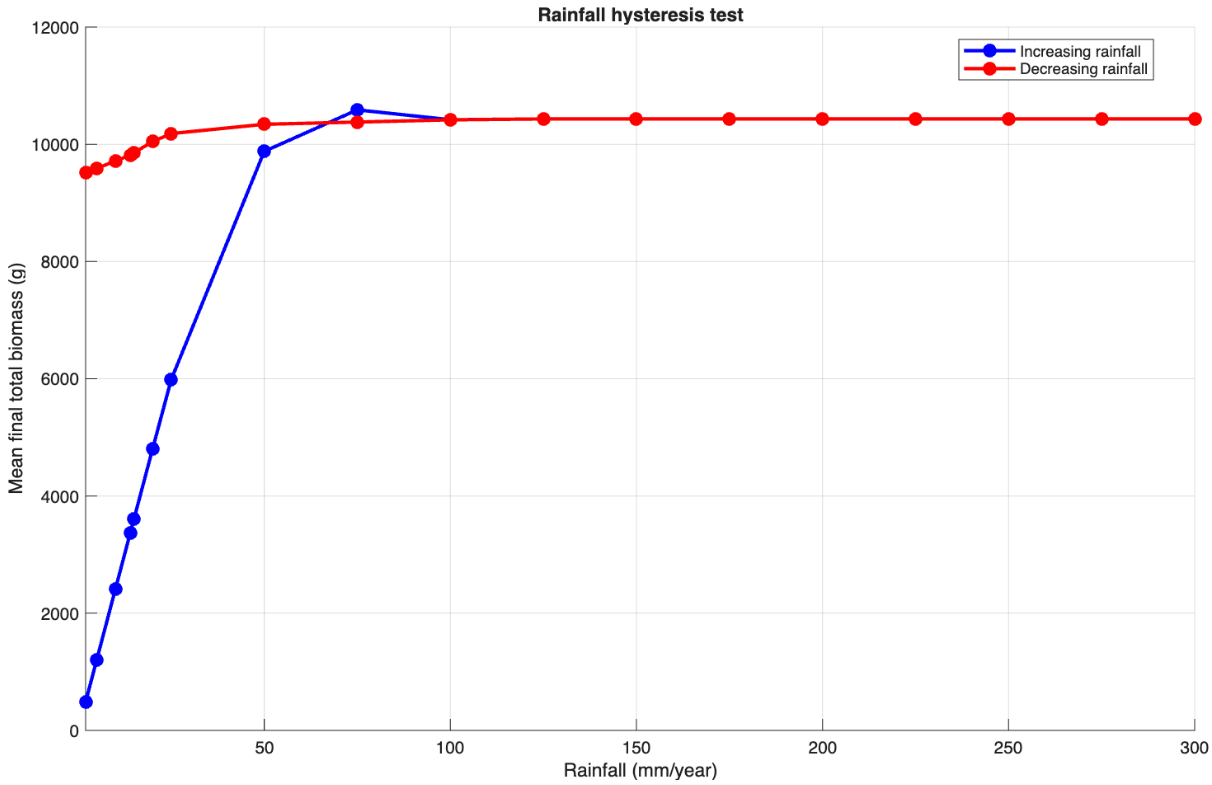


Figure 10: Continuation of the Stewart model, with 500 Year simulations and 0.35gm^{-2} of vertical nitrogen flux.

We see that the model does seem to exhibit some form of hysteresis signature, with the biomass time series data seeming to settle into an oscillatory regime for most runs, but not necessarily a quasi-steady state.

One thing we notice is that sometimes the forward and backward sweeps do not connect, and have small discrepancies. This behaviour seems to happen at random and it is not clear why this is happening. One possible explanation is that the noise in the model is causing the differences, which would make sense as the differences are quite small, almost the size of an oscillation of the biomass. Another explanation is finite simulation time scales not allowing the biomass to converge as much as it could.

4.4.1 Varying Nitrogen

Varying the upwards flux of nitrogen seems to have interesting effects on our continuations.

Generally, lower nitrogen limits growth, and forces the forwards sweep to plateau completely, with only plant maintenance being possible. Note as well water is not at all limiting here as we are increasing the rainfall. Lower nitrogen upwards flux also causes less collapse in the backwards sweep, potentially as the biomass cant actually get that high in the forward sweep because of the lower nitrogen, so the lower rainfall doesn't cause complete collapse as not as much water is required for maintenance of the biomass as the biomass itself is now lower.

Another interesting effect is that lower nitrogen flux seems to flatten oscillations by a considerable amount, but whether or not this has any ecological relevance is unclear, and this is likely just an artefact of the model, due to the plateau of biomass for much of the continuation.

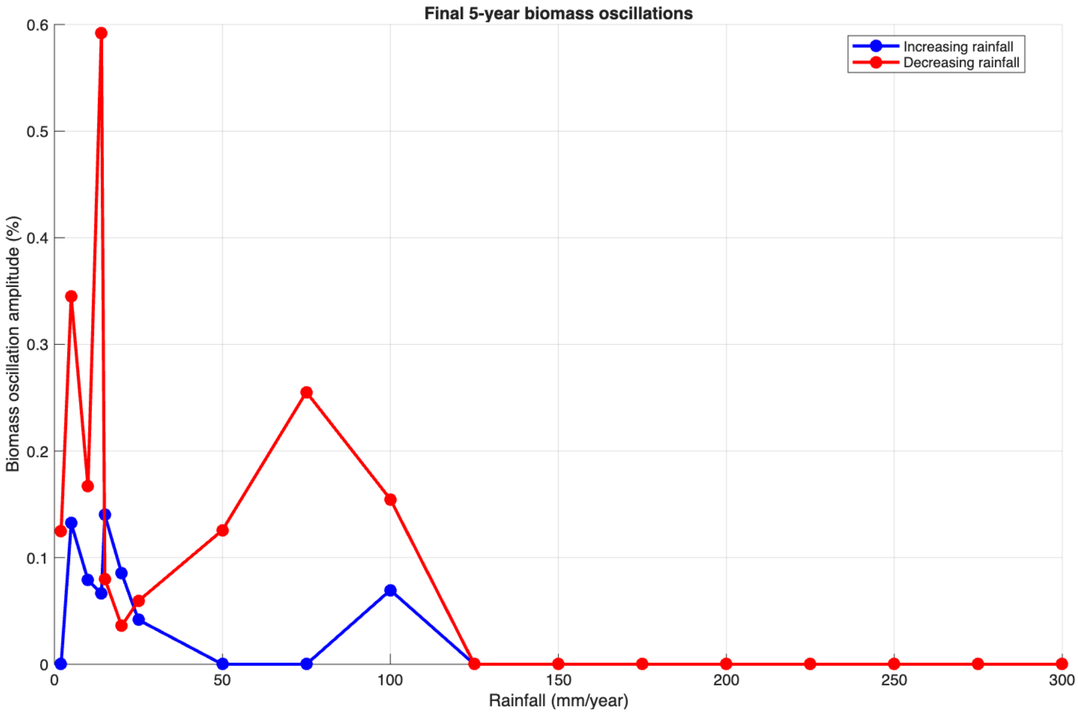
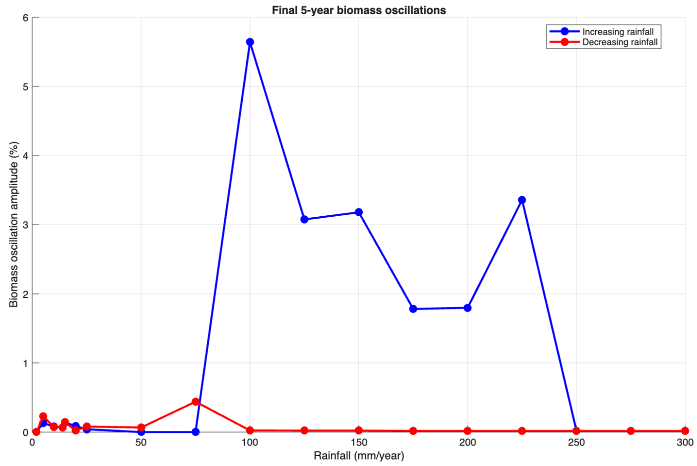


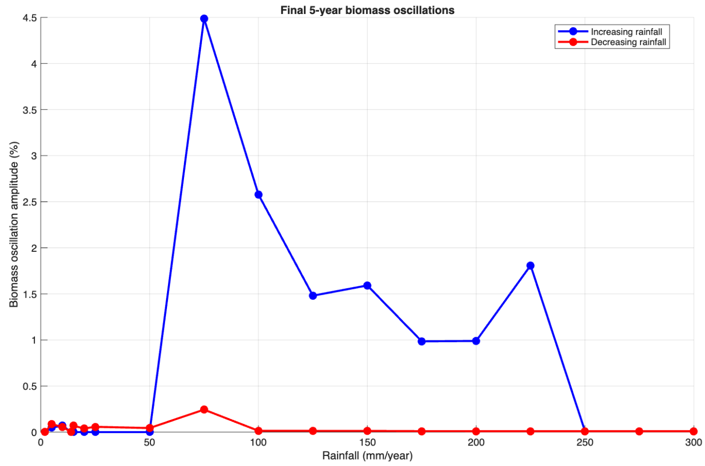
Figure 11: Oscillation amplitude in biomass of the continuation sweep. Measures amplitude of the final 5 years of the simulation with 0.35gm^{-2} of vertical nitrogen flux.

4.4.2 Varying Field Sizes

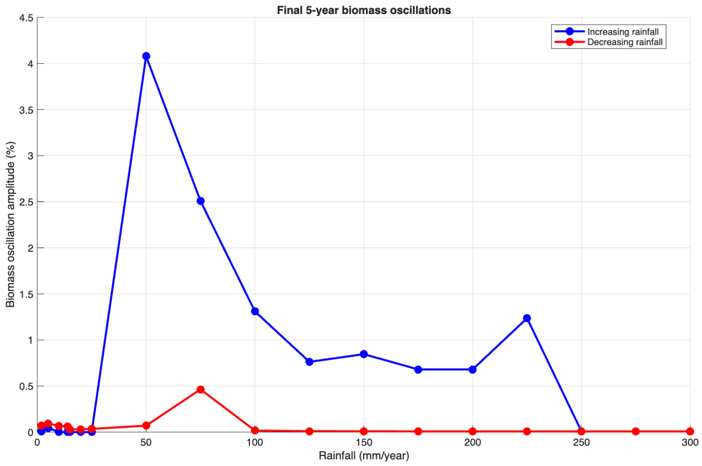
Another interesting investigation is trying to see what happens when we vary field sizes, and whether they produce any substantial changes in the oscillatory behaviour in the continuation branches. In other network models, network size dependence of oscillatory behaviour has been observed [6]. We test the oscillation amplitude by running 500 year simulations of the continuation model, and tracking $\frac{|B_{max}-B_{min}|}{2\overline{B}} \times 100$ in the final 5 years of the simulations. We observe that oscillations generally last 5-10 years.



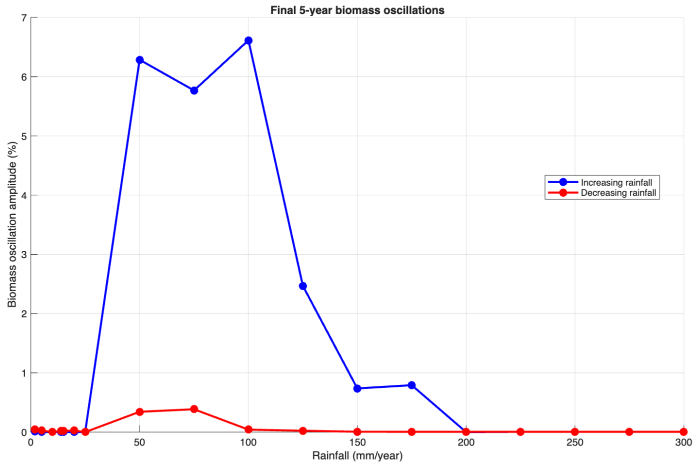
(a) Field size 30×30



(b) Field size 50×50



(c) Field size 70×70



(d) Field size 100×100

Figure 12: Final five-year biomass oscillation amplitudes for different field sizes under increasing and decreasing rainfall continuation. All simulations were ran for 500 years

We find that increasing field sizes does not seem to give a monotonic relationship where increased field size results in smaller amplitude oscillations. The current implementation of the model allows for field sizes up to and including 100×100, meaning that we cannot really test this any further without modifying the subroutines, and furthermore that perhaps $N = 100$ is an edge case. We can see as well that the amplitudes between 75-250mm/year seem to decrease, but then increase again at $N = 100$.

5 Conclusion

Overall, we've studied how the two approaches to modelling vegetation pattern dynamics differ. Mathematically they are substantially different, however both demonstrate similar pattern formation through simple formulations of local resource dynamics, with Klausmeier focussing more on the positive feedback loop between plant uptake and water advection, whereas the Stewart model focuses more on modelling lots of real world forces through discrete time step equations, resulting in patterns.

In the Klausmeier model, we found spatial perturbations of homogeneous equilibria could push the system to instability for ranges of wave numbers, hence generating patterns. Our extensions of Klausmeier allowed us to further control parameters that we use during simulations, allowing us to increase complexity of the underlying assumptions of the model, whilst understanding that this comes with a trade-off of mathematical tractability. We find that Klausmeier's inclusion of just the feedback loop between downslope water advection and plant uptake is enough for vegetation bands, telling us that our extensions are useful in increasing model complexity, but are not essential for pattern formation as expected

Our continuation experiments with the Stewart model produced remarkably different responses and indicated that there are path dependencies in the model, depending on whether we start from low or high rainfall. This behaviour is not too dissimilar from hysteresis, however the oscillations that persist in biomass time series figures tell us that we should be cautious in how we characterise these behaviours. It also gives us a difference in how we should interpret these behaviours. For PDE models, asymptotic steady states are built into the model, and are well represented, including the transitions between them. In the Stewart model, we have transient dynamics where we are in a transition phases of long-term oscillatory regimes rather than fully stationary equilibria.

Overall, whilst for now ignoring the specific issues in the implementations of our models, our comparison boils down to a key trade-off, mathematical tractability, or aligning with the complexity of ecological dynamics, and how can we synthesise these further.

Moving forward, we could investigate why the Stewart model seems to behave in an oscillatory way rather than stationary, and determine whether this is just model noise or the result of model intricacy. We could also investigate further extensions of Klausmeier, and whether or not this can really provide a good intermediate model with more ecological processes included. This could help determine further which ecological processes are actually worth adding to our models, and allow us to continue to build models to predict and control desertification in semi-arid regions.

Code used throughout the project, as well as some sample simulations can be found in

`https://github.com/jianphilpott-source/vegetation-models`.

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